

Jacob Henry Haarberg

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EDUCATION

Washington State University Vancouver

Expected May 2028

B.S. Electrical Engineering (Junior Standing)

Clark College

August 2023 – June 2026

A.S. Electrical Engineering With Honors

Cumulative GPA: 3.6/4.0

- **Full-Time College (2024–2026):** 3.9 GPA | Vice President's List (All Quarters)
- **Running Start (2023–2024):** AA Transfer Coursework (45+ credits completed in High School)

TECHNICAL SKILLS

Hardware: THT/SMD Soldering, Bread/Turret Boards, PIC/Arduino/Pi Microcontrollers

Instrumentation: Oscilloscope, Function Generator, Digital Multimeter, DC/AC Power Supply, LCR meter, ESR meter, Vector Network Analyzer

Software: Spice, Excel, Office, Word, Matlab, Octave, C/C++, PIC18 RISC Assembly, KiCad

CERTIFICATIONS

FCC General Class Amateur Radio License – Call Sign: KM7FWX

PROJECTS

50W High-Voltage Tube Amplifier | *Analog Hardware Design, LTspice, Turret Board* June 2025 – February 2026

Full documentation, schematic files, and build gallery available at: jacobhaarberg.dev

- Simulated the complete amplifier end-to-end in LTspice — power supply rails through signal chain — verifying DC bias, AC response, and stage interactions at the functional block level before committing to hardware
- Translated simulation results into a complete turret board build, producing a fully functional 50W amplifier from hand-drafted layouts
- Engineered a high-voltage linear power supply on turret board delivering 6 output rails (6.3V AC, 270–420V DC, and adjustable -20 to -50V DC bias) using rectification, filter capacitors, filter choke, and negative voltage doubler
- Designed a 5-stage signal path using 5 triodes and 2 pentodes (pre-amp, tone stack, pre-driver, phase inverter, power stage) with both cathode and fixed biasing schemes
- Implemented noise reduction techniques including strategic component placement, transformer orientation to minimize magnetic coupling, and ground loop elimination
- Characterized output transformer impedance via test voltage measurements to achieve proper tube-to-transformer impedance matching

People Counter | *Microcontroller, C++, 3D Printing*

March 2025 – April 2025

- Programmed an Arduino in C++ with a multistage state machine to process ultrasonic sensor data and track bidirectional room occupancy in real time
- Designed and 3D printed a custom 2-digit RGB 7-segment display with individually addressable LEDs, writing driver code to render digits and trigger color-coded maximum occupancy alerts

RELEVANT COURSEWORK

In Progress (Fall 2026): Electronic Devices and Applications (ECE 325), Principles of Solid State Devices (ECE 349), Technical & Professional Writing (ENGLISH 402), Probability & Statistics for Engineers (STATS 360)

Completed: Electrical Circuits (AC/DC), Signals & Systems, Computer Organization & Microprocessors, Engineering Physics I–III, Calculus I–IV, Linear Algebra, Differential Equations

WORK EXPERIENCE

Aquatics Staff Supervisor | *Marshall/Firstenburg Center, Vancouver, WA*

Dec 2025 – Present

Swim Instructor

March 2025 – Dec 2025

Lifeguard

Dec 2024 – Dec 2025

- Supervises a team of 3–5 lifeguards and manages swim lesson execution.
- Measures pool chemistry using titration analysis to maintain safety standards.
- Writes structured, detail-oriented incident and operational reports following department regulations.